

Social Media Sentiment & Customer Engagement Analysis

Project Overview

This project analyzes social media activity to understand **customer sentiment, engagement, platform performance, geographic activity, and time-based patterns**.

The analysis was completed during my **Codveda Technologies Business Analytics Internship** using **Power BI, Power Query, and DAX**.

Business Problem

The company receives a large volume of social media activity across different platforms and markets but needs a clearer understanding of customer sentiment and engagement. Without this visibility, it is difficult to identify high-performing platforms, understand customer reactions, monitor potential sentiment risks, and determine when and where engagement is strongest.

Objectives

The objective was to analyze social media data to identify patterns in:

- Customer sentiment
- Likes and retweets
- Platform performance
- Geographic activity
- Monthly and hourly activity

These findings were then used to develop practical recommendations for improving social media engagement.

Dataset

The dataset contains **732 social media records** with fields including: **Sentiment, Platform, Country, Likes, Retweets, Timestamp, Hashtag, Year, Month, Day, and Hour**.

Tools Used

- **Power BI** — Dashboard and visualization
- **Power Query** — Data cleaning and transformation
- **DAX** — Analytical measures

Analytical Approach

The project followed a structured workflow:

Business Problem → Data Cleaning → Transformation → Analysis → Dashboard → Deeper Analysis → Insights → Recommendations

The analysis focused not only on what was happening in the data, but also on comparing **volume and engagement intensity** to identify more meaningful business insights

Key Results & Business Insights

1. Sentiment & Engagement

Positive was the most frequent sentiment with **45 posts**, but **Joy** generated stronger engagement at approximately **49.3 likes per post**, compared with **37.8 likes per post** for Positive.

Result: The most common sentiment was not necessarily the most engaging.

2. Platform Performance

Instagram recorded the highest activity with **258 posts** and **11,633 likes**. It also had the highest average likes per post at approximately **45.1**.

Result: Instagram showed the strongest overall engagement performance among the three platforms analyzed.

3. Geographic Activity

The **USA** recorded the highest activity with **188 posts** and **8,358 likes**.

Within the USA, Positive had the highest post volume, while Neutral generated higher likes per post.

Result: High activity volume and engagement intensity did not always occur in the same sentiment category.

4. Time-Based Activity

February recorded **85 posts** and **3,505 likes**. Positive led by post volume, while Happiness generated the most likes.

The **2 PM** period recorded the highest hourly activity with **94 posts** and **3,909 likes**, with Joy as the leading sentiment.

Result: Engagement varied across both months and hours, indicating that timing is an important dimension for social media analysis.

Recommendations

- **Prioritize Instagram** for high-impact content.
- **Build on engaging sentiments**, particularly Joy.
- **Monitor sentiment patterns** in high-activity markets such as the USA.
- **Schedule key content around peak periods**, including 2 PM.
- **Track likes and retweets per post** to measure engagement quality over time.

Success Measures

Future performance can be monitored using:

- Likes per post
- Retweets per post
- Sentiment engagement
- Platform engagement
- Peak-period engagement

Conclusion

The analysis shows that social media performance cannot be evaluated by post volume alone.

Sentiment, platform, geography, and timing all provide different perspectives on engagement.

These findings provide a data-driven basis for more targeted content planning, engagement strategies, and ongoing sentiment monitoring.